

From video to Bird View data

Stages 1-2: inspection, frame cutting and pseudo-annotation

1

Bird View inspection and frame cutting

Four development videos were first inspected as aerial Bird View sources. Frames were sampled at a stable cadence and stored with source-video identity, timestamp and split metadata. The held-out **Evaluation** video was not used to create the training data.

SOURCE	FRAMES	ROLE
train A	39	training
train B	63	training
train C	34	training
train D	49	validation
total	185	136 train / 49 validation



Bird View example: dense intersection, scale variation and small road vehicles

2

Pseudo-annotation: two teachers, tiles and merge

YOLO26x and RF-DETR Large were run independently, first on complete frames and then as full-frame-plus-tiles variants. The pilot deliberately measures candidate coverage and teacher agreement, not accuracy: reviewed ground truth did not yet exist at this point.



YOLO26x: 12 frozen pilot frames



RF-DETR Large: same pilot frames

TEACHER / VARIANT	CANDIDATES	CROSS-TEACHER AGREEMENT
YOLO26x, full frame	348	207 same-class matches at IoU 0.50
YOLO26x, full + tiles	543	356 same-class matches at IoU 0.50
RF-DETR Large, full frame	361	pilot comparison
RF-DETR Large, full + tiles	627	small-object recovery; more review work

What entered CVAT

A used the merged YOLO26x + RF-DETR v6 proposals, with an extra RF-DETR high-resolution top ROI at threshold 0.08 to recover missed vehicles. B and C were replaced with RF-DETR 1280-tile proposals. D used a separate RF-DETR deduplicated v3 validation source, not the merged teacher tree.

CVAT review: predictions to labels

Stage 3: manual correction of pseudo-labels on complete source frames

FREQUENT FAILURE MODES AND EXTRA RULES

The pseudo-label stage deliberately retained difficult candidates, so a small rule set prevented obvious geometric failures from overwhelming CVAT review.



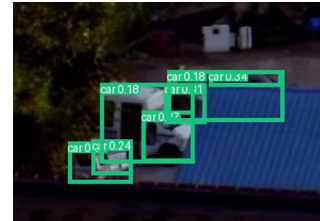
Truck fragments

anchor merge; preserve distinct adjacent trucks



Static-background FP

review building / trailer-like boxes; no blind acceptance



Tile duplicates

class-aware NMS plus containment and centre rules

For elongated trucks, fragment candidates were merged only when containment, longitudinal overlap and anchor-union coverage were consistent; a distinct-neighbour safeguard kept nearby large trucks separate. The train-A high-resolution top ROI additionally used an acceptance polygon, size filters and a lower RF-DETR threshold. Truck was tested as a separate class for a possible distance proxy, but its examples were sparse and inconsistent. The final task therefore uses one *vehicle* class and no distance metric.

3

Manual correction in CVAT

Pseudo-labels were only a starting point. In **CVAT**, each proposed box was reviewed on the full source image: false positives were removed, missed vehicles were added, boundaries were corrected, and all accepted objects were normalised to the one target class, *vehicle*.

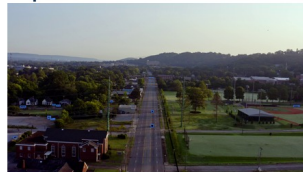
A pseudo-label



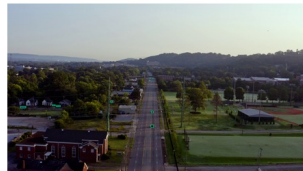
A CVAT reviewed



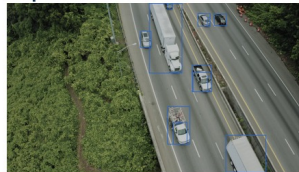
B pseudo-label



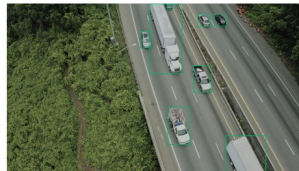
B CVAT reviewed



C pseudo-label



C CVAT reviewed



D pseudo-label



D RF-DETR final



REVIEW-DERIVED PROPOSAL AUDIT

VIDEO	PROPOSED	TP KEPT	FP REMOVED	FN ADDED	FINAL
A	2,903	1,645	1,258	36	1,681
B	1,622	898	724	12	910
C	1,089	893	196	11	904
D	2,769	2,769	0	0	2,769

TP = a proposal matched a final/review box at IoU ≥ 0.50 ; FP = a proposal removed by the final label set; FN = a final/review box without a matching proposal. Boundary-only edits stay matched and are not counted as a separate change.

Important split distinction

A-C are CVAT-reviewed training sources. D is not a manual-CVAT correction set: it is the final RF-DETR deduplicated v3 validation source, so its audit has zero review deltas by construction. This preserves video-level validation separation.

Model adaptation and COCO

Stages 4-5: fit the parameter budget, then learn a vehicle spatial prior

4

YOLO26n model adaptation

The model was adapted step by step to fit the formal 500k-parameter limit without discarding the high-resolution evidence needed for small aerial vehicles. The final graph predicts directly from P2/4, P3/8 and P4/16.

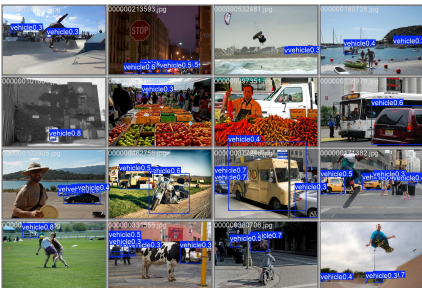
CHANGE	PARAMETERS AFTER	DELTA	WHY IT HELPS
YOLO26n, 80 classes	2,572,280	-	pretrained starting point
One vehicle class	2,504,190	-68,090	match the task taxonomy
Remove extra end-to-end head	2,383,407	-120,783	one post-processing path is sufficient
Remove P5/32, C2PSA and P5 neck/head	574,634	-1,808,773	free budget from the coarsest scale
Narrow new SPPF and P4 neck to 80 ch.	492,794	-81,840	keep compatible backbone capacity
Add direct P2/4 head	492,535	-259	retain detail for tiny vehicles

5

Spatial-perception pre-training on COCO

COCO *car*, *motorcycle*, *bus* and *truck* were remapped to *vehicle*. This provides generic vehicle localisation before fine-tuning on the project camera geometry. The run was first attempted locally on an RTX 3060, but the training time was impractical. It was therefore executed on RunPod L40S with persistent storage; MLflow recorded parameters, metrics and artifacts.

DATA	SETUP	FINAL VALIDATION
18,175 train / 796 val	60 epochs, 960 px, AdamW, batch 32	mAP50 0.632; mAP50-95 0.390



COCO one-class predictions



COCO one-class targets

ONLINE BIRD VIEW AUGMENTATIONS

Frame scale

0.80-2.00x

Horizontal flip

p = 0.50

Mosaic

p = 0.25

Brightness

+/- 0.20

Contrast

+/- 0.20

Saturation

+/- 0.25

Hue

+/- 0.015

Gaussian blur

p = 0.05

JPEG compression

p = 0.10

Bird View training and inference

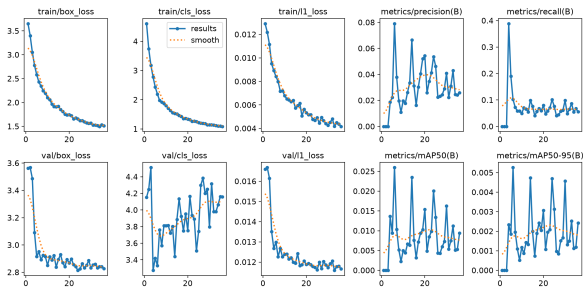
Stages 6-7: aerial fine-tuning, adaptive tiled inference and full-frame output

6

Fine-tuning on the reviewed Bird View dataset

The COCO checkpoint was fine-tuned using A-C for training and D for validation. The planned 120-epoch run ended after 36 epochs; the best D checkpoint was retained. The result is honestly weak: it demonstrates the remaining aerial domain and annotation-quality gap.

AERIAL SPLIT	SELECTED D VALIDATION	INTERPRETATION
136 frames A-C / 49 frames D	best mAP50 0.026; mAP50-95 0.005	not deployment-ready



Aerial learning curves

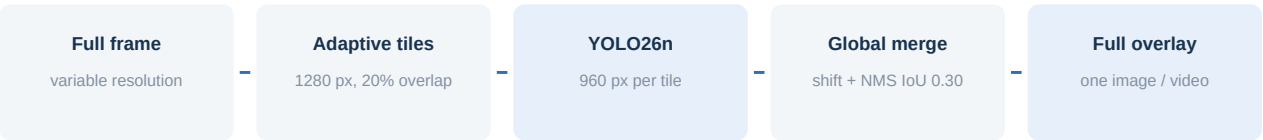


Full-frame inference: train C

7

Adaptive tiled inference and global merging

A 4K frame is not blindly resized before detection: that would make distant vehicles too small. Instead, the frame is divided into overlapping tiles. The tile count is automatically determined by the input resolution, so the same procedure scales to different cameras. If tiles are not desired, the input can be resized directly; scale augmentation during training makes that a supported operating choice.



INFERENCE PARAMETERS	VALUE
tile / overlap	1280 px / 20%
model input	960 px
merge	global class-agnostic NMS, IoU 0.30
rendered output	one full-resolution annotated image per video

Sampled full-frame detections at confidence 0.20: A: 7; B: 6; C: 39; D: 55; Eval: 8. Evaluation remains held out: without reviewed Evaluation ground truth, no held-out mAP or false-alarm rate is claimed.

HELD-OUT TEST-FRAME BEHAVIOUR



The held-out Evaluation video differs sharply from the Bird View training distribution: nearby vehicles occupy much larger image regions. The small-object-trained detector therefore produces false positives and poorly sized boxes on this frame.

This visual result is intentionally reported as a limitation, not a success metric. It motivates a more diverse reviewed training set and a separate validation protocol for large, near-camera vehicles.